

# Smaller and Shallower XOR Circuits for AES MixColumns: 97 Gates at Depth 3; 92 at Depth 4; 89 at Depth 5

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## Abstract

We give four verified implementations of AES MixColumns as circuits of 2-input XOR gates over  $\text{GF}(2)$ . Three improve the published depth–count Pareto frontier at their depth: **97** gates at depth **3**, the minimum depth, against 99 [2]; **92** at depth **4** against 97 [3]; **89** at depth **5** against 94 [3]. The fourth, **88** at depth **7**, *ties* the published gate-count floor of Jean [9] — who has priority — with an independent circuit sharing 61 of 88 internal masks. All were found by stochastic local search over a value-set formulation of the shortest-linear-program problem. No 87 was found: 47 canonical 88-gate circuits have exhaustively empty remove- $\leq 3$  neighbourhoods. Nothing here is claimed optimal.

## 1 Model and specification

A circuit is an ordered list of 2-input XOR gates:  $s_0, \dots, s_{31}$  are the input bits, and gate  $k$  produces  $s_{32+k} = s_a \oplus s_b$  with  $a, b < 32 + k$ . Cost is the gate count; depth counts gates on the longest input-to-output path, inputs at depth 0. Bit index  $i$  denotes bit  $(i \bmod 8)$  of state byte  $\lfloor i/8 \rfloor$ , least-significant-bit first, and MixColumns maps an input column  $a_0, \dots, a_3$  to

$$\text{out}[c] = 2 \cdot a_c \oplus 3 \cdot a_{(c+1) \bmod 4} \oplus a_{(c+2) \bmod 4} \oplus a_{(c+3) \bmod 4},$$

in  $\text{GF}(2^8)$  modulo  $x^8 + x^4 + x^3 + x + 1$  (0x11b), per NIST FIPS 197-upd1, Eq. 5.6 [1]. The map is  $\text{GF}(2)$ -linear, so agreement on the 32 unit-input vectors is a complete correctness check; counts and depths are relabeling-invariant, so no comparison below depends on convention.

The s-XOR (in-place) model is comparable: SSA-unrolling a  $k$ -instruction s-XOR program yields exactly  $k$  free-standing gates. Multi-input XOR gates, gate-equivalent area and quantum CNOT counts are different cost functions: not comparable, not claimed against.

## 2 Results

Depth 3 is the known minimum depth — an output depending on  $w$  inputs needs depth  $\geq \lceil \log_2 w \rceil$ , and MixColumns has outputs of weight 7 [2] — so the depth-3 circuit’s contribution is its count. The 89 at depth 5 is shallower than any published circuit of fewer than 94 gates whose depth is stated, and stays on the frontier: the published sub-89 point, Jean’s 88 [9], measures depth 7 on our transcription — that note states no depth, exactly as [8] does not.

The 88 at depth 7 *ties* Jean’s point rather than improving it: Jean’s note was posted 2026-07-23 and has priority. Ours was found 2026-07-26 on a fully logged own lineage — from-scratch 97 at depth 3 (235s), a 21 h ladder to 89 at depth 6 (11.0h), one reroute to 89 at depth 5 (592s),  $\rho^2$ -symmetrization to a 94 at depth 5, then 32.9 min of plateau walking

| Depth          | Published best | Model        | This work         | Relation                                 |
|----------------|----------------|--------------|-------------------|------------------------------------------|
| 3              | 99 [2]         | XOR2         | <b>97</b>         | −2 (depth 3 is the minimum depth)        |
| 4              | 97 [3]         | XOR2         | <b>92</b>         | −5                                       |
| 5              | 94 [3]         | XOR2         | <b>89</b>         | −5                                       |
| 6              | 92 [4, 5]      | XOR2 / s-XOR | —                 | dominated by 89 @ 5                      |
| 7 <sup>‡</sup> | 88 [9]         | XOR2         | <b>88</b>         | ties it; independent, 61/88 masks shared |
| 8              | —              | XOR2         | (88) <sup>†</sup> | derived from [9]; dominated by 88 @ 7    |
| 9 <sup>‡</sup> | 89 [8]         | g-XOR        | —                 | off the frontier                         |

Table 1: The published depth–count landscape in comparable models, and this work. <sup>†</sup>Derived and non-frontier: listed for completeness, claimed as nothing. <sup>‡</sup>*Neither* [9] *nor* [8] states a depth: the 7 and the 9 are both our own measurements of our own transcriptions, on the same footing. Jean’s 88 [9] supersedes the earlier 91 of [6, 7] at depth 7; the depth-3 lineage is 105 → 103 → 102 → 99 ([2], Table 3). Listings: Appendices A–E.

— with no imported circuit in the chain, and is a different circuit: 61 of 88 masks shared, Jaccard 0.530. The fifth circuit, 88 at depth 8 (Appendix E), is derived work and dominated: its seed chain passes through Jean’s,  $\rho^2$ -symmetrized and peeled to 95, orbit-walked to 92, then unioned with a 91 of our own lineage before the descent.

### 3 Method

A circuit is represented as a *value set*: masks over the 32 inputs, valid iff every mask is the XOR of two already-available masks and all 32 output masks are present; gate count is the set size. Deletion becomes a set operation, circuits compare by mask overlap regardless of gate order, and a neighbourhood — remove  $k$  masks, add at most  $k - 1$  — becomes finite and enumerable (Section 4).

Good circuits at this size *look* locally efficient — every gate reads as an optimal step — and that is what makes them hard to improve: they sit at local optima, so smaller circuits are separated from them by locally *suboptimal* steps. (This was true of the naive construction we started from and of every circuit we produced up to the 89; we make no claim about how anyone else’s circuit was built.) **Remove-1** drops a non-output mask if the set stays realizable. The **neutral swap** removes one mask and adds one repair mask at unchanged size: with  $A = \text{closure}(S')$  and  $\text{stuck} = (S' \cup T) \setminus A$ , every single-mask repair lies in  $C = \{u \oplus a : u \in \text{stuck}, a \in A\} \cap P_2(A)$  ( $P_2 = \text{pair sums}$ ), so enumerating  $C$  *decides* the move — measured  $|C| \approx 7$ , cheaper than the earlier engine’s 24–40 random candidates, which missed 81–82% of existing repairs. **Remove-2-add-1** remains a diversifier, though at the 88/89 frontier it is provably empty (Section 4). **Destroy-and-rebuild** removes a group and re-synthesizes from a pool, accepting rebuilds no larger, or larger on an annealing schedule with reheat; destroying a *connected cone* of the DAG leaves a hole worth re-planning, where  $k$  unrelated masks are usually just re-derived, and near-miss rebuilds are peeled before judgement. A fixpoint over minimum build depths rejects cap violations and emits gate lists in depth order.

Workers run one engine at one depth cap in fixed-length chunks, alternating walk and rebuild, and a coordinator re-offers each the best circuit feasible at its cap (the reseeding ladder). A worker accepts fewer gates *or equal gates at strictly lower depth* — a Pareto tie-break, and not a cosmetic one: 89 at depth 6 became 89 at depth 5 through it, and both 88s, first seen at depths 11 and 10, reached depths 7 and 8 seconds later. Every candidate is verified against MixColumns rebuilt from  $\text{GF}(2^8)$  before acceptance. The depth-3 circuit comes from a separate constructor: each output is modelled as  $A \oplus B$ ,  $A$  and  $B$  partitioning its bits into shared, refcount-costed depth- $\leq 2$  parts, annealed by iterated local search, so

depth  $\leq 3$  holds by construction.

**The v2 engine.** Version 1’s circuits came from an engine whose kernel was then profiled and rewritten [10]: level-BFS depth relaxation ( $17.5\times$  on destroy-and-rebuild, bit-identical trajectories); a worklist closure with incremental removal queries, and removal of a bug that re-verified unchanged circuits every plateau iteration — 78 % of walk runtime ( $21.9\times$  on the walk); victim repooling ( $\sim 34\times$  accepted moves/s); cone destroys ( $\sim 12\times$ ); exact repair ( $\sim 7\times$  plateau mobility). The walk now runs at 480–640 iterations/s against 70–110. What found both 88s was not a cleverer move but a cheap plateau walked at volume: every distinct equal-best mask set is harvested.

## 4 Exact local certificates, and structure

To say what is *not* there we decide the neighbourhood question: given a verified 88-gate circuit and a window of  $k$  removed masks, can at most  $k - 1$  new masks — each buildable when added, with cascade unlocks between — restore all 32 targets? A “yes” is an 87-gate circuit; a completed “no” is a proof for that window. Completeness follows from a first-unlock case analysis: the first unlock must have the newest added mask as a parent, which pins each added mask to a finite, enumerable shape. Runs above the proved budget are recorded as undecided, never as irreducible. The procedure agrees with an independent brute force 25/25 and 12/12 at  $k \leq 3$ , and on 1 257 instances including 122 genuine negatives at  $k = 4$ .

- **47 canonical 88-gate circuits** — Jean’s 88, its 12 plateau siblings, and a third family’s anchor with 33 representatives — **have exhaustively empty  $k \leq 3$  shells**: all 1 540 windows at  $k = 2$  and all 27 720 at  $k = 3$  each, none reducible. Hence *any 87-gate circuit differs from each of the 47 by at least 4 masks*. Stated plainly: all 47 lie in Jean’s-lineage families — families 1–2 are Jean’s circuit and its siblings, and the third family’s anchor is itself seeded through Jean’s circuit — so this is a rigidity statement about *that* neighbourhood, and the one circuit here independent of that lineage, our own 88 at depth 7, is exactly the least-certified: it is *not* among the 47, its exhaustive  $k \leq 3$  sweep was never run, and it has only 9 exact  $k = 4$  windows and 8 SAT cone windows. The best-certified shell and the independent shell are disjoint.
- **Population and depth census.** Harvesting mapped  $\approx 139\,878$  distinct 88-gate mask sets, of which 105 801 are proven irreducible at  $k = 2$ . Every one has minimum build depth  $\geq 7$ , and 18 353 (13.1 %) reach depth exactly 7: no 88-gate circuit of depth  $\leq 6$  has been seen (depths recomputed over the whole harvest, itself listed by size and SHA-256 in [10]). In total  $\approx 165$  million exact window decisions, zero timeouts, zero reducible windows; all five 89-gate seeds likewise have empty  $k = 2$  shells.
- **Windowed SAT is evidence, not proof.** A sound windowed CNF encoding, solved by Kissat under two soundness-proved symmetry-breaking layers, returned UNSAT up to  $k = 16$  on Jean’s 88 and  $k = 15$  on the third-family anchor (whose whole  $k \leq 12$  regime is UNSAT, 28/28), with **0 SAT anywhere** — but UNSAT there is relative to the encoding’s fixed slot order, and undecided windows are undecided.

None of this bounds 87 away globally; it is consistent with 88 being locally rigid, and it redirected the search: at 88 gates every move class small enough to enumerate is empty, so what remains is walking the equal-size plateau widely.

**Symmetry.** Byte rotation  $\rho$  (order 4) commutes with MixColumns, so the 32 targets form 8  $\rho$ -orbits of size 4 and equivariant circuits can be searched in orbit space — but full  $\rho$ -symmetry costs about +19 gates. The half-rotation  $\rho^2$  is where the structure is: in the elite basin 79–81 % of a circuit’s masks have their  $\rho^2$ -image also in the mask set, and the 12  $\rho^2$ -fixed masks the search settles on are exactly the classic  $x_0 \oplus x_2$  /  $x_1 \oplus x_3$  pairs, rediscovered rather than imposed. Jean’s 88, at 75 % (66 of 88), symmetrizes into a second, distinct basin. A  $\rho^2$ -equivariant orbit engine produced two *exactly*  $\rho^2$ -symmetric 90-gate circuits (depths 9 and 7), improving the best exactly symmetric circuit we know of from 94 to 90; both are certified locally optimal in orbit space, and both are derived work, each seed union containing a circuit symmetrized from Jean’s 88. Symmetry is also a seeding device: the exactly  $\rho^2$ -symmetric 94 at depth 5 — 82 of its 94 masks shared with the 89 at depth 5 — costs +5 over that 89 and is the seed from which the 88 at depth 7 was found.

## 5 Verification, artifacts, and reproduction

Every circuit here is checkable from the note alone: the appendices list every gate, Section 1 defines the target map, and because MixColumns is linear, replaying a listing on the 32 unit-input vectors against Eq. 5.6 [1] is a complete correctness check. Each appendix also states its circuit’s canonical SHA-256 over the UTF-8 bytes of `{"inputCount":32,"gates":[...]}`.

The artifacts automate that. Circuits and verifiers are at <https://github.com/Joe-b-20/aes-mixcolumns-xor-circuits>, archived at concept DOI 10.5281/zenodo.21299092; `python3 verify_all.py` runs `verify.py`, which rebuilds the target map from the  $\text{GF}(2^8)$  definition so the verifier *is* the specification, a clean-room verifier with adversarial rejection tests, and Verilog testbenches, all under continuous integration. The method, pipeline, untouched run archives with each run’s exact code, certificate summaries and the credited transcriptions of the imported circuits [9, 8] are published separately [10], one reproduction command per result. Measured there on 2026-07-27 with the shipped v2 engine, and reported as measurements rather than promises: the from-scratch 97 at depth 3 in 81 s on one core; the 89 at depth 5 in 19 s and 22 s from shipped seeds, against 592 s for the archived 2026-07-14 run on the v1 engine; the 88 at depth 7 in 19.4 and 31.0 minutes in two single-worker re-runs from the  $\rho^2$ -symmetric 94, against 32.9 min in the archived ten-worker run; the 92 at depth 4 a from-scratch cascade of hours. The re-runs are re-runs, not independent confirmations.

Implementation and execution used LLM coding agents as programming tools, directed by the author, who designed the moves and acceptance rules. Concretely: the three frontier records of this note are reproduced by the published, dependency-free code with no AI system in the loop, whereas the v2 engine, the certificates of Section 4 and both 88-gate circuits came out of a 24-agent LLM campaign over 2026-07-26/27, again author-directed. That distinction changes nothing about correctness: every circuit any of it produced is machine-verified against MixColumns rebuilt from  $\text{GF}(2^8)$ , and the method is fully specified and reproducible from the accompanying code either way.

## 6 Scope and claims

- **Not optimality.** Minimum 2-input-XOR circuit size (the Shortest Linear Program problem) is NP-hard; no count here is proven minimal, and the certificates of Section 4 are local.
- **Comparisons.** All counts are for the model and convention of Section 1; the baselines [2, 3, 4, 5, 6, 7, 8, 9] were source-checked, and a source-by-source audit, with a dated Corrections section, is in the artifact repository’s `PRIOR_ART.md`. We claim the

smallest counts at depths 3, 4 and 5, and no improvement at unconstrained depth: the published floor is 88 and stays 88.

- **Priority.** The 88 at depth 7 ties [9], posted first; the 88 at depth 8 is derived from that circuit and dominated. Neither is a record claim. The depths quoted for [9] and [8] — 7 and 9 — are our measurements of our own transcriptions; neither paper states a depth.
- **Earlier results.** This note supersedes the project’s 98 at depth 3, 91 at depth 6 and 89 at depth 10, kept in the artifact repository.

**Version history.** **v2**, 27 July 2026: adds the 88 at depth 7 and its derived depth-8 companion, the exact neighbourhood certificates, the depth census and  $\rho^2$  symmetry results, the v2 engine measurements and dated reproduction times; no claim of v1 is withdrawn. **v1**, July 2026, archived at version DOI 10.5281/zenodo.21299093: the 97, 92 and 89 only.

## References

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- [2] H. Shi, X. Feng, and S. Xu, *A Framework with Improved Heuristics to Optimize Low-Latency Implementations of Linear Layers*, IACR Transactions on Symmetric Cryptology, 2023(4):489–510, 2023. <https://doi.org/10.46586/tosc.v2023.i4.489-510>
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- [10] J. Bachir, *The method: value-set SLP search, plateau walking, and exact local certificates* (METHODS.md, evidence/RESULTS.md, reproduce/), slp-plateau-search v2.0.0, 2026. <https://github.com/Joe-b-20/slp-plateau-search>

## A Circuit mixcolumns\_97gates\_depth3 (97 gates, depth 3)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

e4c4e678c4b7c22d889c6e060b43859c111b82b953d6d9b43c49c079c0545b56

|                        |                        |                         |                         |
|------------------------|------------------------|-------------------------|-------------------------|
| s32 = s0 $\oplus$ s8   | s57 = s7 $\oplus$ s31  | s82 = s51 $\oplus$ s41  | s107 = s75 $\oplus$ s18 |
| s33 = s5 $\oplus$ s13  | s58 = s23 $\oplus$ s31 | s83 = s53 $\oplus$ s3   | s108 = s80 $\oplus$ s66 |
| s34 = s6 $\oplus$ s14  | s59 = s33 $\oplus$ s14 | s84 = s52 $\oplus$ s40  | s109 = s84 $\oplus$ s66 |
| s35 = s7 $\oplus$ s15  | s60 = s35 $\oplus$ s17 | s85 = s52 $\oplus$ s21  | s110 = s62 $\oplus$ s55 |
| s36 = s8 $\oplus$ s16  | s61 = s39 $\oplus$ s10 | s86 = s42 $\oplus$ s54  | s111 = s86 $\oplus$ s89 |
| s37 = s9 $\oplus$ s17  | s62 = s41 $\oplus$ s5  | s87 = s53 $\oplus$ s54  | s112 = s93 $\oplus$ s64 |
| s38 = s10 $\oplus$ s18 | s63 = s35 $\oplus$ s40 | s88 = s43 $\oplus$ s56  | s113 = s67 $\oplus$ s58 |
| s39 = s11 $\oplus$ s19 | s64 = s42 $\oplus$ s7  | s89 = s55 $\oplus$ s30  | s114 = s94 $\oplus$ s72 |
| s40 = s11 $\oplus$ s20 | s65 = s36 $\oplus$ s43 | s90 = s57 $\oplus$ s0   | s115 = s75 $\oplus$ s71 |
| s41 = s12 $\oplus$ s20 | s66 = s43 $\oplus$ s19 | s91 = s57 $\oplus$ s3   | s116 = s95 $\oplus$ s79 |
| s42 = s6 $\oplus$ s22  | s67 = s32 $\oplus$ s24 | s92 = s57 $\oplus$ s15  | s117 = s96 $\oplus$ s81 |
| s43 = s15 $\oplus$ s23 | s68 = s36 $\oplus$ s24 | s93 = s34 $\oplus$ s58  | s118 = s87 $\oplus$ s62 |
| s44 = s1 $\oplus$ s24  | s69 = s44 $\oplus$ s37 | s94 = s44 $\oplus$ s58  | s119 = s89 $\oplus$ s34 |
| s45 = s1 $\oplus$ s25  | s70 = s32 $\oplus$ s46 | s95 = s48 $\oplus$ s58  | s120 = s92 $\oplus$ s56 |
| s46 = s9 $\oplus$ s25  | s71 = s46 $\oplus$ s10 | s96 = s53 $\oplus$ s58  | s121 = s90 $\oplus$ s36 |
| s47 = s2 $\oplus$ s26  | s72 = s46 $\oplus$ s16 | s97 = s68 $\oplus$ s35  | s122 = s69 $\oplus$ s90 |
| s48 = s11 $\oplus$ s26 | s73 = s45 $\oplus$ s17 | s98 = s70 $\oplus$ s60  | s123 = s74 $\oplus$ s2  |
| s49 = s2 $\oplus$ s27  | s74 = s45 $\oplus$ s38 | s99 = s74 $\oplus$ s77  | s124 = s76 $\oplus$ s91 |
| s50 = s3 $\oplus$ s27  | s75 = s47 $\oplus$ s37 | s100 = s78 $\oplus$ s61 | s125 = s82 $\oplus$ s91 |
| s51 = s4 $\oplus$ s27  | s76 = s47 $\oplus$ s39 | s101 = s63 $\oplus$ s83 | s126 = s85 $\oplus$ s33 |
| s52 = s4 $\oplus$ s28  | s77 = s46 $\oplus$ s26 | s102 = s87 $\oplus$ s85 | s127 = s86 $\oplus$ s59 |
| s53 = s12 $\oplus$ s28 | s78 = s49 $\oplus$ s35 | s103 = s59 $\oplus$ s56 | s128 = s88 $\oplus$ s64 |
| s54 = s13 $\oplus$ s29 | s79 = s50 $\oplus$ s18 | s104 = s93 $\oplus$ s15 |                         |
| s55 = s21 $\oplus$ s29 | s80 = s50 $\oplus$ s38 | s105 = s65 $\oplus$ s67 |                         |
| s56 = s22 $\oplus$ s30 | s81 = s51 $\oplus$ s19 | s106 = s65 $\oplus$ s73 |                         |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128]

## B Circuit mixcolumns\_92gates\_depth4 (92 gates, depth 4)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

e79bfccf5322fe8933242b47b53e94e73163615c70d5c45c9721e27ac86fd97d

|                        |                        |                         |                          |
|------------------------|------------------------|-------------------------|--------------------------|
| s32 = s10 $\oplus$ s26 | s56 = s15 $\oplus$ s31 | s80 = s45 $\oplus$ s49  | s104 = s63 $\oplus$ s49  |
| s33 = s12 $\oplus$ s20 | s57 = s43 $\oplus$ s44 | s81 = s21 $\oplus$ s33  | s105 = s64 $\oplus$ s50  |
| s34 = s14 $\oplus$ s30 | s58 = s34 $\oplus$ s53 | s82 = s67 $\oplus$ s81  | s106 = s64 $\oplus$ s69  |
| s35 = s6 $\oplus$ s14  | s59 = s5 $\oplus$ s54  | s83 = s57 $\oplus$ s66  | s107 = s23 $\oplus$ s75  |
| s36 = s11 $\oplus$ s28 | s60 = s0 $\oplus$ s37  | s84 = s58 $\oplus$ s77  | s108 = s71 $\oplus$ s80  |
| s37 = s8 $\oplus$ s16  | s61 = s38 $\oplus$ s52 | s85 = s42 $\oplus$ s81  | s109 = s46 $\oplus$ s72  |
| s38 = s7 $\oplus$ s31  | s62 = s6 $\oplus$ s40  | s86 = s59 $\oplus$ s67  | s110 = s102 $\oplus$ s66 |
| s39 = s17 $\oplus$ s25 | s63 = s32 $\oplus$ s56 | s87 = s59 $\oplus$ s40  | s111 = s18 $\oplus$ s101 |
| s40 = s13 $\oplus$ s21 | s64 = s36 $\oplus$ s56 | s88 = s10 $\oplus$ s73  | s112 = s89 $\oplus$ s78  |
| s41 = s2 $\oplus$ s26  | s65 = s37 $\oplus$ s56 | s89 = s45 $\oplus$ s79  | s113 = s63 $\oplus$ s108 |
| s42 = s5 $\oplus$ s29  | s66 = s1 $\oplus$ s48  | s90 = s23 $\oplus$ s68  | s114 = s108 $\oplus$ s55 |
| s43 = s16 $\oplus$ s24 | s67 = s50 $\oplus$ s53 | s91 = s62 $\oplus$ s52  | s115 = s93 $\oplus$ s65  |
| s44 = s9 $\oplus$ s25  | s68 = s35 $\oplus$ s56 | s92 = s58 $\oplus$ s62  | s116 = s47 $\oplus$ s109 |
| s45 = s18 $\oplus$ s26 | s69 = s3 $\oplus$ s38  | s93 = s60 $\oplus$ s76  | s117 = s52 $\oplus$ s107 |
| s46 = s0 $\oplus$ s17  | s70 = s51 $\oplus$ s54 | s94 = s35 $\oplus$ s77  | s118 = s96 $\oplus$ s104 |
| s47 = s1 $\oplus$ s9   | s71 = s19 $\oplus$ s48 | s95 = s15 $\oplus$ s61  | s119 = s33 $\oplus$ s106 |
| s48 = s23 $\oplus$ s31 | s72 = s24 $\oplus$ s38 | s96 = s69 $\oplus$ s74  | s120 = s71 $\oplus$ s105 |
| s49 = s3 $\oplus$ s27  | s73 = s39 $\oplus$ s41 | s97 = s65 $\oplus$ s72  | s121 = s103 $\oplus$ s51 |
| s50 = s4 $\oplus$ s20  | s74 = s41 $\oplus$ s55 | s98 = s38 $\oplus$ s60  | s122 = s97 $\oplus$ s100 |
| s51 = s12 $\oplus$ s27 | s75 = s34 $\oplus$ s38 | s99 = s70 $\oplus$ s71  | s123 = s90 $\oplus$ s75  |
| s52 = s22 $\oplus$ s30 | s76 = s43 $\oplus$ s48 | s100 = s57 $\oplus$ s46 |                          |
| s53 = s13 $\oplus$ s29 | s77 = s22 $\oplus$ s42 | s101 = s39 $\oplus$ s78 |                          |
| s54 = s4 $\oplus$ s28  | s78 = s41 $\oplus$ s44 | s102 = s39 $\oplus$ s65 |                          |
| s55 = s11 $\oplus$ s19 | s79 = s10 $\oplus$ s47 | s103 = s69 $\oplus$ s50 |                          |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[97, 122, 89, 118, 119, 82, 84, 90, 115, 110, 111, 113, 120, 85, 91, 117, 93, 83, 88, 114, 99, 86, 92, 95, 98, 116, 112, 96, 121, 87, 94, 123]

## C Circuit mixcolumns\_89gates\_depth5 (89 gates, depth 5)

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

c3b972af282d25e8313a510f5cf840472401c6268bd28cda889593fc3379f064

|                        |                        |                         |                           |
|------------------------|------------------------|-------------------------|---------------------------|
| s32 = s9 $\oplus$ s17  | s55 = s21 $\oplus$ s39 | s78 = s55 $\oplus$ s66  | s101 = s60 $\oplus$ s50   |
| s33 = s19 $\oplus$ s27 | s56 = s5 $\oplus$ s46  | s79 = s37 $\oplus$ s57  | s102 = s39 $\oplus$ s101  |
| s34 = s1 $\oplus$ s25  | s57 = s14 $\oplus$ s42 | s80 = s35 $\oplus$ s56  | s103 = s34 $\oplus$ s87   |
| s35 = s21 $\oplus$ s29 | s58 = s15 $\oplus$ s37 | s81 = s56 $\oplus$ s66  | s104 = s59 $\oplus$ s89   |
| s36 = s12 $\oplus$ s28 | s59 = s11 $\oplus$ s33 | s82 = s55 $\oplus$ s42  | s105 = s76 $\oplus$ s70   |
| s37 = s22 $\oplus$ s30 | s60 = s11 $\oplus$ s43 | s83 = s70 $\oplus$ s72  | s106 = s92 $\oplus$ s65   |
| s38 = s3 $\oplus$ s27  | s61 = s0 $\oplus$ s52  | s84 = s43 $\oplus$ s71  | s107 = s96 $\oplus$ s98   |
| s39 = s4 $\oplus$ s28  | s62 = s33 $\oplus$ s45 | s85 = s54 $\oplus$ s67  | s108 = s36 $\oplus$ s93   |
| s40 = s9 $\oplus$ s25  | s63 = s38 $\oplus$ s52 | s86 = s54 $\oplus$ s49  | s109 = s95 $\oplus$ s48   |
| s41 = s13 $\oplus$ s29 | s64 = s43 $\oplus$ s51 | s87 = s17 $\oplus$ s64  | s110 = s1 $\oplus$ s92    |
| s42 = s5 $\oplus$ s13  | s65 = s8 $\oplus$ s45  | s88 = s57 $\oplus$ s67  | s111 = s99 $\oplus$ s46   |
| s43 = s15 $\oplus$ s23 | s66 = s36 $\oplus$ s41 | s89 = s60 $\oplus$ s47  | s112 = s98 $\oplus$ s75   |
| s44 = s6 $\oplus$ s22  | s67 = s41 $\oplus$ s44 | s90 = s62 $\oplus$ s53  | s113 = s89 $\oplus$ s90   |
| s45 = s7 $\oplus$ s23  | s68 = s34 $\oplus$ s53 | s91 = s71 $\oplus$ s74  | s114 = s108 $\oplus$ s99  |
| s46 = s12 $\oplus$ s20 | s69 = s24 $\oplus$ s45 | s92 = s24 $\oplus$ s61  | s115 = s110 $\oplus$ s112 |
| s47 = s10 $\oplus$ s18 | s70 = s40 $\oplus$ s47 | s93 = s38 $\oplus$ s62  | s116 = s3 $\oplus$ s104   |
| s48 = s2 $\oplus$ s26  | s71 = s31 $\oplus$ s49 | s94 = s58 $\oplus$ s52  | s117 = s90 $\oplus$ s109  |
| s49 = s6 $\oplus$ s14  | s72 = s26 $\oplus$ s34 | s95 = s59 $\oplus$ s63  | s118 = s108 $\oplus$ s101 |
| s50 = s19 $\oplus$ s20 | s73 = s32 $\oplus$ s48 | s96 = s64 $\oplus$ s69  | s119 = s87 $\oplus$ s112  |
| s51 = s8 $\oplus$ s16  | s74 = s44 $\oplus$ s45 | s97 = s61 $\oplus$ s51  | s120 = s32 $\oplus$ s110  |
| s52 = s7 $\oplus$ s31  | s75 = s40 $\oplus$ s51 | s98 = s0 $\oplus$ s65   |                           |
| s53 = s2 $\oplus$ s18  | s76 = s18 $\oplus$ s73 | s99 = s4 $\oplus$ s63   |                           |
| s54 = s30 $\oplus$ s35 | s77 = s10 $\oplus$ s68 | s100 = s58 $\oplus$ s74 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[96, 119, 83, 113, 118, 78, 79, 84, 107, 103, 76, 116, 102, 80, 85, 91, 106, 115, 105, 117, 114, 81, 86, 94, 97, 120, 77, 109, 111, 82, 88, 100]

## D Circuit mixcolumns\_88gates\_depth7 (88 gates, depth 7; ties the published floor)

*Status: the same (depth, count) point as Jean [9], reached independently on this project's own lineage; 61 of 88 internal masks are shared. It ties that point and does not beat it, and Jean has priority.*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

76bac50ab0e4a7dd54578e77f57f6d08e96a8803aff72dba2c3f5fadaaa06ee1

|                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|
| s32 = s13 $\oplus$ s29 | s43 = s5 $\oplus$ s13  | s54 = s33 $\oplus$ s30 | s65 = s43 $\oplus$ s63 |
| s33 = s21 $\oplus$ s29 | s44 = s15 $\oplus$ s31 | s55 = s34 $\oplus$ s15 | s66 = s44 $\oplus$ s45 |
| s34 = s22 $\oplus$ s30 | s45 = s10 $\oplus$ s26 | s56 = s35 $\oplus$ s20 | s67 = s45 $\oplus$ s58 |
| s35 = s15 $\oplus$ s23 | s46 = s6 $\oplus$ s22  | s57 = s35 $\oplus$ s8  | s68 = s46 $\oplus$ s49 |
| s36 = s17 $\oplus$ s25 | s47 = s6 $\oplus$ s14  | s58 = s36 $\oplus$ s2  | s69 = s47 $\oplus$ s54 |
| s37 = s12 $\oplus$ s20 | s48 = s11 $\oplus$ s19 | s59 = s37 $\oplus$ s5  | s70 = s47 $\oplus$ s31 |
| s38 = s12 $\oplus$ s28 | s49 = s7 $\oplus$ s23  | s60 = s39 $\oplus$ s1  | s71 = s48 $\oplus$ s56 |
| s39 = s9 $\oplus$ s25  | s50 = s7 $\oplus$ s31  | s61 = s39 $\oplus$ s51 | s72 = s49 $\oplus$ s19 |
| s40 = s2 $\oplus$ s18  | s51 = s18 $\oplus$ s26 | s62 = s41 $\oplus$ s50 | s73 = s49 $\oplus$ s16 |
| s41 = s3 $\oplus$ s27  | s52 = s32 $\oplus$ s46 | s63 = s42 $\oplus$ s21 | s74 = s50 $\oplus$ s24 |
| s42 = s4 $\oplus$ s28  | s53 = s32 $\oplus$ s38 | s64 = s43 $\oplus$ s14 | s75 = s50 $\oplus$ s55 |

|                        |                        |                           |                           |
|------------------------|------------------------|---------------------------|---------------------------|
| s76 = s51 $\oplus$ s72 | s88 = s64 $\oplus$ s34 | s100 = s86 $\oplus$ s48   | s112 = s104 $\oplus$ s87  |
| s77 = s52 $\oplus$ s54 | s89 = s66 $\oplus$ s86 | s101 = s87 $\oplus$ s37   | s113 = s104 $\oplus$ s71  |
| s78 = s52 $\oplus$ s64 | s90 = s67 $\oplus$ s85 | s102 = s90 $\oplus$ s61   | s114 = s106 $\oplus$ s108 |
| s79 = s53 $\oplus$ s63 | s91 = s68 $\oplus$ s70 | s103 = s90 $\oplus$ s40   | s115 = s107 $\oplus$ s74  |
| s80 = s53 $\oplus$ s59 | s92 = s70 $\oplus$ s35 | s104 = s94 $\oplus$ s38   | s116 = s107 $\oplus$ s110 |
| s81 = s55 $\oplus$ s68 | s93 = s71 $\oplus$ s42 | s105 = s94 $\oplus$ s40   | s117 = s107 $\oplus$ s57  |
| s82 = s57 $\oplus$ s16 | s94 = s72 $\oplus$ s3  | s106 = s95 $\oplus$ s60   | s118 = s111 $\oplus$ s41  |
| s83 = s58 $\oplus$ s61 | s95 = s73 $\oplus$ s74 | s107 = s96 $\oplus$ s44   | s119 = s111 $\oplus$ s66  |
| s84 = s59 $\oplus$ s33 | s96 = s73 $\oplus$ s0  | s108 = s96 $\oplus$ s36   |                           |
| s85 = s60 $\oplus$ s17 | s97 = s82 $\oplus$ s9  | s109 = s97 $\oplus$ s108  |                           |
| s86 = s62 $\oplus$ s76 | s98 = s82 $\oplus$ s24 | s110 = s98 $\oplus$ s49   |                           |
| s87 = s62 $\oplus$ s4  | s99 = s85 $\oplus$ s97 | s111 = s100 $\oplus$ s105 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[110, 109, 103, 119, 113, 79, 88, 92, 115, 99, 83, 89, 93, 84, 77, 91, 116, 106, 67, 100, 112, 80, 69, 75, 117, 114, 102, 118, 101, 65, 78, 81]

## E Circuit mixcolumns\_88gates\_depth8 (88 gates, depth 8; derived, non-frontier)

*Status: derived from published work, and not a frontier point. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), and it is dominated by Appendix D (same count, greater depth). It is listed as a verified distinct construction, not as a claim.*

Signals  $s_0, \dots, s_{31}$  are the inputs; gate  $k$  produces  $s_{32+k}$ . Canonical gate-list SHA-256 (see Section 5):

3a23e2f2b2f3835cab74ba23800088bb03ab4ee3f076f2b9a826c78cd84f8ef1

|                        |                        |                         |                           |
|------------------------|------------------------|-------------------------|---------------------------|
| s32 = s13 $\oplus$ s29 | s55 = s35 $\oplus$ s8  | s78 = s59 $\oplus$ s75  | s101 = s95 $\oplus$ s20   |
| s33 = s21 $\oplus$ s29 | s56 = s38 $\oplus$ s26 | s79 = s59 $\oplus$ s63  | s102 = s96 $\oplus$ s16   |
| s34 = s22 $\oplus$ s30 | s57 = s39 $\oplus$ s50 | s80 = s60 $\oplus$ s34  | s103 = s96 $\oplus$ s8    |
| s35 = s15 $\oplus$ s23 | s58 = s40 $\oplus$ s21 | s81 = s61 $\oplus$ s41  | s104 = s97 $\oplus$ s36   |
| s36 = s9 $\oplus$ s17  | s59 = s42 $\oplus$ s45 | s82 = s63 $\oplus$ s76  | s105 = s100 $\oplus$ s89  |
| s37 = s12 $\oplus$ s28 | s60 = s43 $\oplus$ s14 | s83 = s64 $\oplus$ s66  | s106 = s100 $\oplus$ s39  |
| s38 = s2 $\oplus$ s18  | s61 = s43 $\oplus$ s58 | s84 = s66 $\oplus$ s35  | s107 = s101 $\oplus$ s94  |
| s39 = s3 $\oplus$ s27  | s62 = s45 $\oplus$ s19 | s85 = s67 $\oplus$ s19  | s108 = s101 $\oplus$ s37  |
| s40 = s4 $\oplus$ s28  | s63 = s46 $\oplus$ s26 | s86 = s67 $\oplus$ s11  | s109 = s102 $\oplus$ s44  |
| s41 = s4 $\oplus$ s20  | s64 = s47 $\oplus$ s49 | s87 = s68 $\oplus$ s0   | s110 = s102 $\oplus$ s68  |
| s42 = s9 $\oplus$ s25  | s65 = s48 $\oplus$ s53 | s88 = s74 $\oplus$ s17  | s111 = s106 $\oplus$ s86  |
| s43 = s5 $\oplus$ s13  | s66 = s48 $\oplus$ s31 | s89 = s76 $\oplus$ s44  | s112 = s106 $\oplus$ s38  |
| s44 = s15 $\oplus$ s31 | s67 = s49 $\oplus$ s3  | s90 = s77 $\oplus$ s41  | s113 = s109 $\oplus$ s68  |
| s45 = s10 $\oplus$ s18 | s68 = s50 $\oplus$ s24 | s91 = s81 $\oplus$ s33  | s114 = s110 $\oplus$ s42  |
| s46 = s1 $\oplus$ s25  | s69 = s50 $\oplus$ s54 | s92 = s81 $\oplus$ s51  | s115 = s112 $\oplus$ s93  |
| s47 = s6 $\oplus$ s22  | s70 = s51 $\oplus$ s58 | s93 = s85 $\oplus$ s89  | s116 = s113 $\oplus$ s103 |
| s48 = s6 $\oplus$ s14  | s71 = s52 $\oplus$ s53 | s94 = s85 $\oplus$ s40  | s117 = s113 $\oplus$ s55  |
| s49 = s7 $\oplus$ s23  | s72 = s52 $\oplus$ s60 | s95 = s86 $\oplus$ s35  | s118 = s114 $\oplus$ s97  |
| s50 = s7 $\oplus$ s31  | s73 = s54 $\oplus$ s64 | s96 = s87 $\oplus$ s49  | s119 = s114 $\oplus$ s88  |
| s51 = s32 $\oplus$ s37 | s74 = s55 $\oplus$ s16 | s97 = s87 $\oplus$ s1   |                           |
| s52 = s32 $\oplus$ s47 | s75 = s56 $\oplus$ s36 | s98 = s88 $\oplus$ s46  |                           |
| s53 = s33 $\oplus$ s30 | s76 = s56 $\oplus$ s10 | s99 = s94 $\oplus$ s77  |                           |
| s54 = s34 $\oplus$ s15 | s77 = s57 $\oplus$ s12 | s100 = s95 $\oplus$ s62 |                           |

Output signals (output bit  $j$  is carried by the listed signal,  $j = 0, \dots, 31$ ):

[116, 119, 79, 112, 108, 70, 80, 84, 109, 98, 75, 111, 107, 92, 71, 83, 103, 118, 78, 115, 99, 91, 65, 69, 117, 104, 82, 105, 90, 61, 72, 73]